

Rutu Patel

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Professional Summary

Machine Learning Engineer skilled in Computer Vision, NLP, Data Analysis, and IoT-based intelligent systems with hands-on experience in YOLOv8, OpenCV, Flask, and Scikit-learn. Developed real-time drone detection systems and AI-powered legal judgment analysis platforms using research-driven Machine Learning approaches. Strong foundation in feature engineering, model evaluation, object detection, explainable AI, and end-to-end ML pipeline development.

Technical Skills

- **Languages:** Python, SQL
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Computer Vision, NLP, Classification, Regression, Model Evaluation
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, Seaborn, Flask
- **Computer Vision:** YOLOv8, Object Detection, Motion Analysis, Real-Time Detection Systems
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Power BI, ESP32, Telegram API

Experience

AI/ML Engineer Intern

Jan 2026 – Apr 2026

VishlRAFrosy.ai

Remote

- Developed and evaluated ML models using Python and Scikit-learn for predictive analytics on real-world datasets.
- Performed data preprocessing, feature engineering, model training, hyperparameter optimization, and performance evaluation.
- Contributed to AI research workflows and intelligent analytical solution development.

React.js Frontend Intern

Dec 2024 – Jan 2025

SoftCoding Solution

- Built responsive frontend interfaces and reusable UI components using React.js.
- Integrated REST APIs and optimized application performance and user experience.

Projects

Physics-Informed IoT-Based Drone Detection System

Python, YOLOv8, OpenCV, ESP32, Telegram API

[GitHub Repository: github.com/Rutu0912/DroneDetection](#)

- Developed a real-time drone detection system using Deep Learning and Computer Vision achieving reliable distinction between drones and birds.
- Implemented physics-informed filtering (trajectory, speed, acceleration, stability) on top of YOLOv8 to reduce false positives significantly.
- Designed a complete AI + IoT pipeline integrating ESP32 and Telegram API for automated real-time alert generation.
- Evaluated system performance on 10+ real-world drone videos and custom datasets.

AI-Assisted Legal Judgment Analysis System

Python, NLP, Flask, TF-IDF, Cosine Similarity, Indian Kanoon API

[GitHub Repository: github.com/Rutu0912/AIJudicialReason](#)

- Developed an AI-powered legal judgment analysis system for Company, Criminal, and Property Law domains using NLP techniques.
- Performed preprocessing and TF-IDF-based feature extraction on legal documents and used cosine similarity for relevant case retrieval.
- Integrated Indian Kanoon API to access legal judgments and precedent cases.
- Built a Flask web application for legal domain classification and structured judgment analysis.

Education

Aditya Silver Oak Institute of Technology

2022 – 2026

Bachelor of Technology (B.Tech) in Computer Engineering | CGPA: 9.07/10

R.P.P Girls High School & S.G Shah Higher Secondary School

Secondary & Higher Secondary Education (Science Stream)